Can Lab Measurements Predict Good Wine?

Classifying Vinho Verde quality from physicochemical properties

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1. Introduction

Portugal's Vinho Verde region certifies its wines with laboratory tests that measure each batch's physicochemical properties, such as acidity, sugar, sulfur dioxide and alcohol. Each wine is also scored for quality by a panel of expert tasters. Cortez et al. (2009) collected these records and used regression models to predict the tasters' 0–10 score. I asked a simpler question: **can the lab measurements alone tell a good wine from an ordinary one?**

To answer it, I turned the score into a binary label. A wine is *good* if it scored 7 or higher, and *not good* otherwise. I chose 7 as the cutoff because very few wines score 8 or above, and a higher cutoff would leave too few good wines to learn from. I then compared nine models on held-out data, covering linear, nonparametric, tree-based and kernel methods.

2. Data

The data contain 1,599 red and 4,898 white wines. Each has 11 predictors: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free and total sulfur dioxide, density, pH, sulphates and alcohol. I modeled red and white wines separately because their chemistry differs systematically. I split each data set at random into a 70% training set and a 30% test set.

Two features of the data shape everything that follows:

- **The classes are imbalanced.** Only 14% of red wines and 22% of white wines are good. As a result, a "model" that labels every wine *not good* already has a test error of 0.142 on red and 0.207 on white. Any useful model has to beat this baseline, and it also has to identify the good wines. For that reason I report **sensitivity** (the share of good wines a model catches) alongside the error rate.
- **The classes overlap heavily.** Good and ordinary wines have similar chemistry, so no decision boundary separates them cleanly.

3. Methods

I fit the models in three stages, and each stage built on what the previous one revealed.

1. **K-nearest neighbors, LDA and QDA.** These three are standard starting points that span a range of flexibility. For KNN, I chose k by leave-one-out cross-validation on the training set. I ran KNN twice: once on the raw measurements and once on standardized ones.
2. **Classification trees.** I fit an unpruned tree, then a pruned tree whose size was chosen by 10-fold cross-validation, and finally a bagged ensemble of 500 trees.
3. **Support vector machines** with a radial kernel. I fit one SVM on the two predictors the trees identified as most important, and another on all 11. In both cases I tuned the cost parameter by 10-fold cross-validation.

I scored every model on the test set, which none of the models saw during fitting or tuning.

4. Results

Model	Red error	Red sens.	White error	White sens.
Baseline: always “not good”	0.142	0%	0.207	0%
KNN (raw features)	0.127	51%	0.195	55%
KNN (standardized)	0.117	62%	0.156	64%
LDA	0.110	41%	0.192	31%
QDA	0.140	49%	0.236	69%
Tree (unpruned)	0.110	40%	0.213	44%
Tree (pruned)	0.119	41%	0.213	44%
Bagged trees	0.092	54%	0.127	55%
SVM (alcohol + volatile acidity)	0.142	0%	0.193	23%
SVM (all predictors)	0.096	49%	0.170	34%

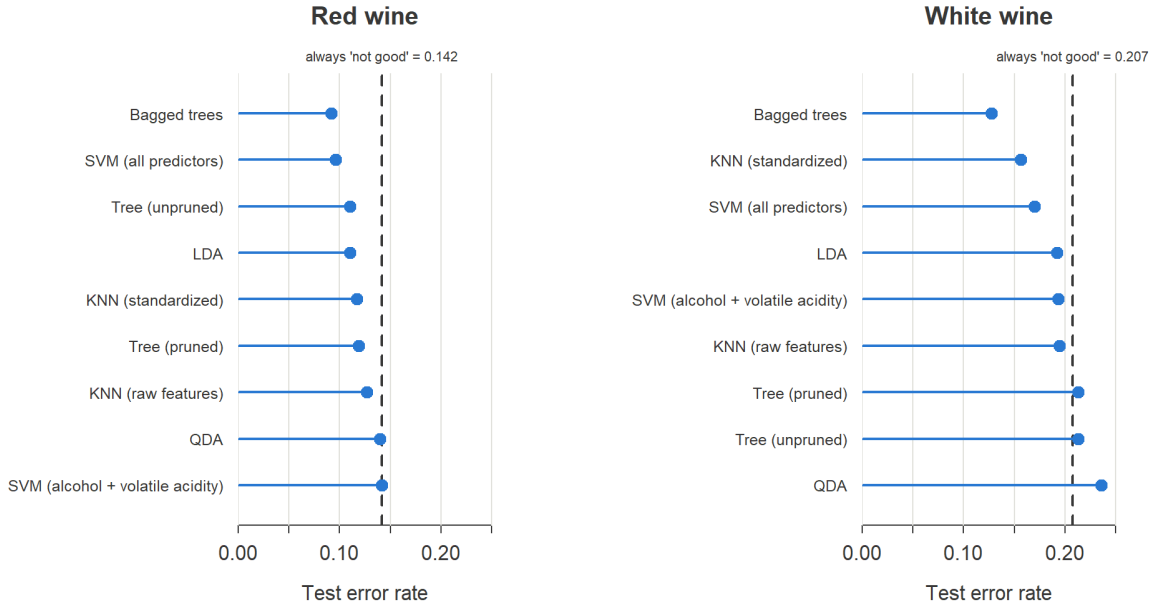


Figure 1: Test error rate by model. The dashed line marks the error from always predicting “not good”; points to its left beat the baseline.

4.1. KNN, LDA and QDA

KNN’s weakness in this problem turned out to be scale rather than dimensionality. The sulfur dioxide measurements run into the hundreds, while most other predictors are below 1. On raw features, the distances KNN relies on are therefore driven almost entirely by sulfur dioxide. Standardizing the predictors cut the white wine test error from 0.195 to 0.156, and the red error from 0.127 to 0.117.

LDA was the strongest of the three on red wine, with a test error of 0.110. That figure looks good, but it hides a problem: the model caught only 41% of the good wines. Its low error comes mostly from correctly labeling the majority class. QDA did worse than LDA on both wines, and on white wine it did worse than the baseline. QDA estimates a separate covariance matrix for each class, which is 66 parameters per class here. With only 149 good red wines in the training set, those estimates are noisy. Several predictors are also heavily skewed, which violates the normality assumption QDA depends on more heavily than LDA does.

The LDA discriminant plots show the underlying difficulty. Good wines shift to the right of ordinary ones, but the two distributions overlap substantially.

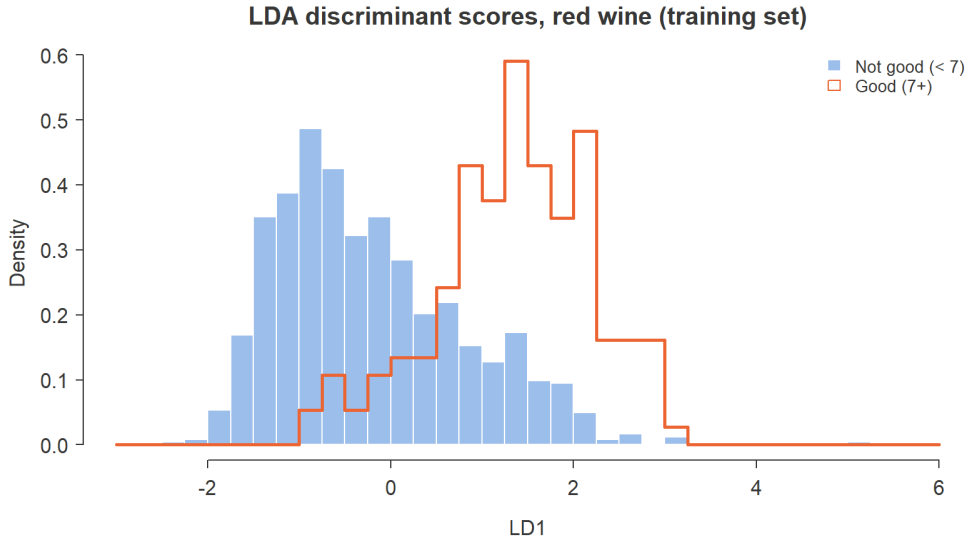


Figure 2: LDA discriminant scores on the training set, red wine.

4.2. Classification trees

The trees showed that **alcohol** is by far the most informative predictor in both wines. The first split in each tree is on alcohol content. In red wine, the next most useful predictors are volatile acidity and sulphates. The unpruned red tree had 18 leaves, while the white tree had only 5. Few of the possible splits in white wine reduced deviance enough to pass the tree's default stopping rule, which is another sign of how much the classes overlap.

Pruning chose 7 leaves for red wine and 3 for white. It made the trees much easier to read but did not improve accuracy: red test error rose slightly, from 0.110 to 0.119, and white was unchanged. Bagging was the clear improvement. It reduced test error to 0.092 for red and 0.127 for white, the best results in the study. Averaging over 500 bootstrapped trees lowers the high variance that makes a single tree unstable.

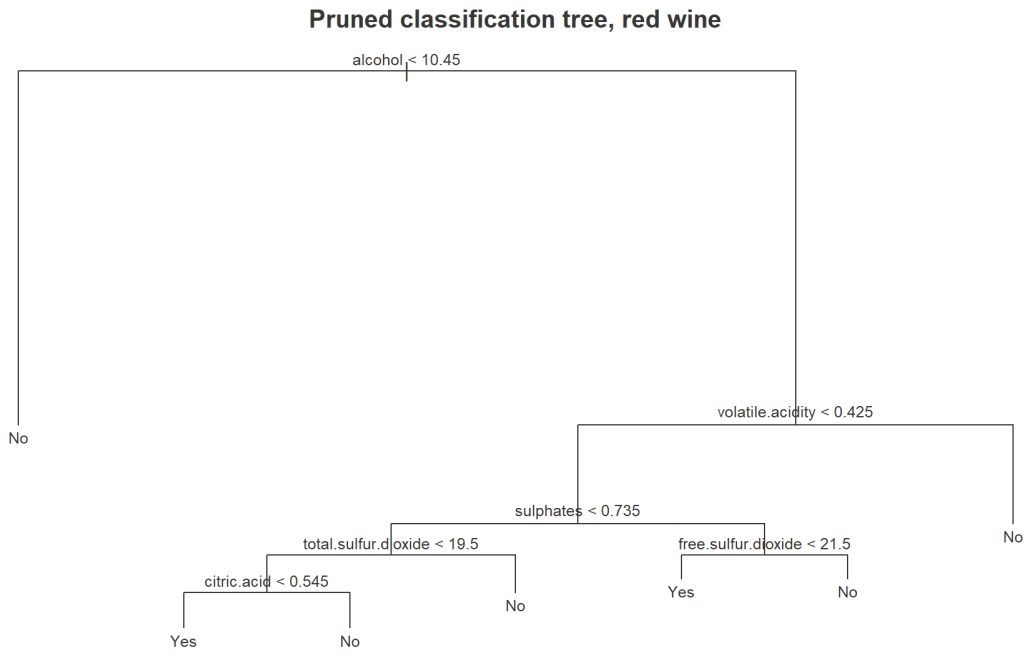


Figure 3: Pruned classification tree, red wine.

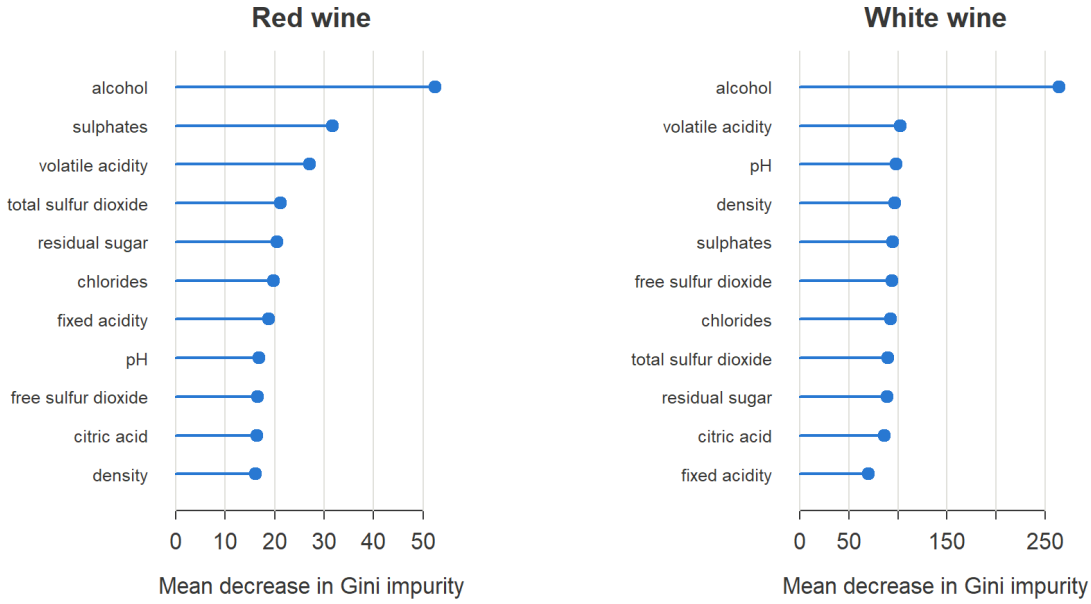


Figure 4: Variable importance from the bagged trees (mean decrease in Gini impurity).

4.3. Support vector machines

Because the trees singled out alcohol and volatile acidity, I first fit an SVM on just those two predictors. It failed. On red wine, cross-validation chose the smallest cost in the grid, and the resulting model predicted *not good* for every wine, so its error equals the baseline. On white wine it beat the baseline only slightly and caught 23% of the good wines. The decision region plot shows why. In these two dimensions, good and ordinary wines are interleaved. The white wine model needed 1,489 support vectors out of 3,428 training wines, which means a large share of the data sits on or inside the margin.

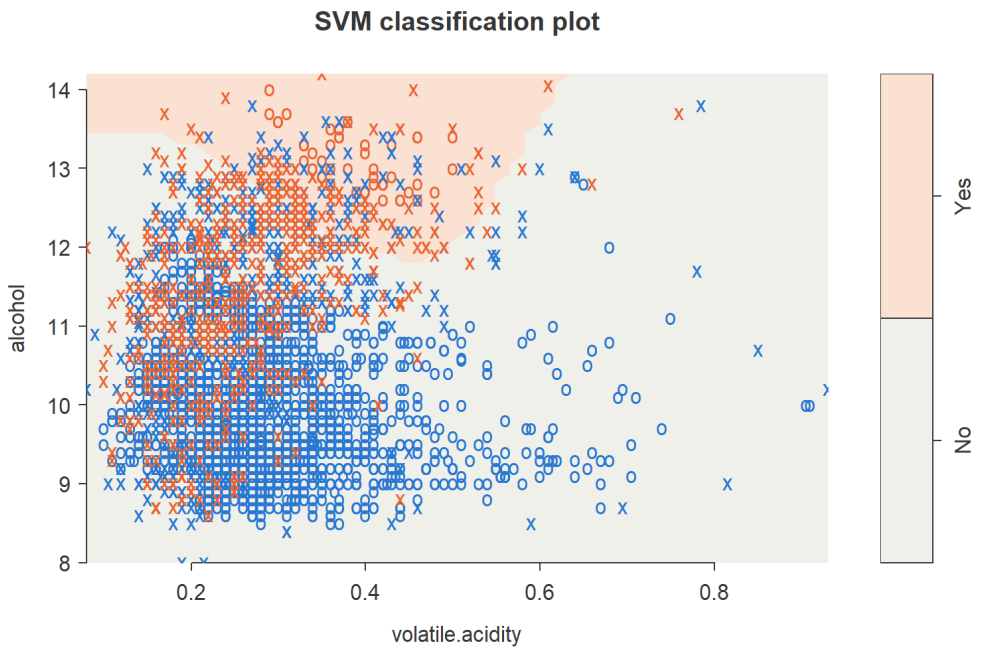


Figure 5: Two-predictor SVM on white wine. Shaded regions are the predicted class, orange marks are good wines, and × marks are support vectors.

Using all 11 predictors helped considerably. Test error fell to 0.096 for red, nearly matching bagging, and to 0.170 for white. The two strongest predictors carry much of the signal, but the other nine still add information.

5. Discussion

Lab measurements can predict wine quality, but only partly, and the error rate alone overstates how well. The bagged trees performed best on both wines. Their error was about a third below the baseline on red and close to 40% below it on white. Even so, they caught only about half of the good wines. For a winery deciding which batches to promote, sensitivity matters as much as overall accuracy, and the models are noticeably weaker on that measure.

White wine was harder to classify across almost every method. The size of the data set is not the reason; more observations do not make a classifier less accurate. Part of the gap is mechanical. Good wines make up 22% of white wines but only 14% of reds, so even the baseline error is higher for white. The rest reflects greater overlap between the classes in white wine, which shows in how few useful splits the white tree could find.

I also considered raising the cutoff to 8, but too few wines would qualify as good, and the class imbalance would become much worse. Applying class weights or choosing a different classification threshold would be a more direct way to trade a little overall accuracy for higher sensitivity.

Cortez et al. treated quality as a numeric score and used regression, with support vector regression giving their best results. Their approach keeps information that a binary label discards. A wine scoring 6 and one scoring 3 are both *not good* to my classifiers, but they are very different wines. Ordinal logistic

regression would be a natural middle ground. It respects the ordering of the scores, and it stays simple and interpretable, with one set of coefficients and a series of cutpoints between adjacent quality levels.

Finally, the response variable itself limits how well any model can do. Each wine's score is the median of at least three blind tastings, and tasters judge aroma, appearance and taste. None of these are recorded directly in the lab data. The two strongest predictors do have plausible sensory explanations. Higher alcohol gives a wine more body and warmth. Volatile acidity, which is mostly acetic acid, gives a vinegar-like off-note at high levels and is treated as a fault. Still, much of what a taster responds to is not captured by these 11 measurements. A different panel could also score the same wine differently, so some irreducible error is unavoidable.

5.1. Limitations

- **Duplicate records.** About 15% of red wines and 19% of white wines have predictor values identical to another wine. Some duplicates fall in both the training and test sets, which flatters methods that can memorize individual points, such as KNN with $k = 1$ and bagged trees. Removing duplicates before splitting would give a more conservative estimate.
- **One train/test split.** All results come from a single random split. Repeated cross-validation would show how much the ranking of models depends on which wines happen to fall in the test set.

6. Conclusion

I compared nine classification models for predicting whether a Vinho Verde wine is rated good from its lab measurements. Bagged trees performed best, with SVMs on all 11 predictors close behind for red wine. Alcohol content was by far the most important predictor. However, because the classes are imbalanced and overlap heavily, even the best models identify only about half of the good wines, and several popular methods do no better than always guessing *not good*. The most important lesson of the project is to compare every model against that baseline, and to measure how well it finds the minority class, before trusting its accuracy.

7. References

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems, 47*(4), 547–553. https://doi.org/10.1016/j.dss.2009.05.016

Cortez, P., Teixeira, J., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Using data mining for wine quality assessment. In *Discovery Science* (Lecture Notes in Computer Science, Vol. 5808, pp. 66–79). Springer. https://doi.org/10.1007/978-3-642-04747-3_8

Originally written as the final project for STAT 7510; revised in 2026. Code and data: see the repository README.